RISE

IONIZING RADIATION: EXPERIMENTAL SESSION

Project implemented with the support of the Ministry of Education and Research - UEFISCDI, project number PN-IV-P10-SS-SC-2024-0189, funded through PNCDI IV

The RISE project (Ionizing Radiation - Experimental Session) aims to help understand the science of ionizing radiation in schools through interactive experimental and educational activities.

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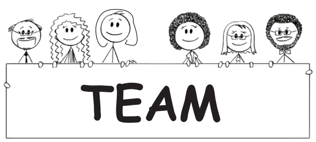

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Contents

Ionizing Radiation

Ionizing radiation is all around us. It occurs naturally in our environment and is also produced for a wide range of practical applications. It plays an essential role in medicine, industry, scientific research, and energy production. At the same time, excessive exposure can be harmful to living organisms, making it important to understand where ionizing radiation comes from, how it interacts with matter, and how it can be used safely.

This chapter introduces the main types of ionizing radiation, where they are encountered in everyday life, and why they can be both beneficial and potentially hazardous. It provides the foundation needed to understand the experiments and measurements presented later in this manual.

What is Ionizing Radiation?

Ionizing radiation consists of particles or electromagnetic waves that carry enough energy to remove electrons from atoms or molecules. This process is known as ionization. When ionization occurs, an atom or molecule loses an electron and becomes an electrically charged ion.

The most common types of ionizing radiation are **alpha radiation, beta radiation**, and **X** and **gamma rays**..

ALPHA RADIATION

Alpha radiation consists of helium nuclei (having two protons and two neutrons) traveling at very high speeds. Because alpha particles carry a positive electric charge, they interact with the electrons in the materials they pass through. These frequent interactions produce many ionizations, causing the particles to lose their energy rapidly. As a result, alpha radiation has very little penetrating power. It can be stopped by just a few centimeters of air or even a single sheet of paper.

BETA RADIATION

In most cases, beta radiation consists of high-energy electrons. These electrons are approximately **7000 times lighter** and **10 times faster** than alpha particles.

Like alpha particles, electrons are electrically charged. However, their much lower mass and higher speed make them more difficult to stop. Beta radiation can typically be shielded by a few millimeters of aluminum, plastic, or other lightweight materials.

X AND GAMMA RAYS

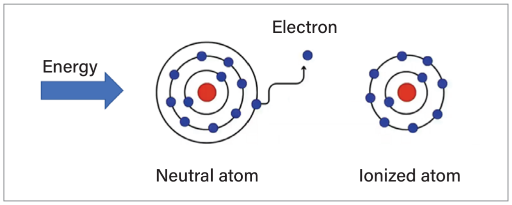
X and gamma rays are physically identical; the different names originate from historical convention. Usually the distinction is based on how they are produced: X rays originate from interactions involving electrons, whereas gamma rays are emitted by atomic nuclei.

Although X and gamma rays are electromagnetic waves, they often behave as discrete particles called photons, illustrating the principle of wave-particle duality. A photon of visible light typically has an energy of about 2–3 eV, whereas photons of X and gamma rays have energies on the order of thousands or even millions of electronvolts.

Unlike alpha and beta particles, photons carry no electric charge and can pass through matter without interacting. When an interaction does occur however, a photon transfers part or all of its energy to a particle in the material - most often an electron - producing ionization.

Since these interactions occur only with a certain probability rather than every time a photon passes through matter, X and gamma rays are far more penetrating than alpha or beta radiation and can travel through substantial thicknesses of material. It takes dense materials such as lead to greatly reduce their intensity.

Type of Radiation	Electric Charge	Penetrating Power
Alpha	Positive	Low
Beta	Negative	Moderate
X / Gamma rays	None	High



Ionization is a fundamental process by which radiation transfers energy to matter. Depending on the application, this process can be beneficial - for example sterilizing medical equipment or treating certain types of cancer. In other situations, ionization can damage healthy cells, which is why excessive exposure to ionizing radiation must be controlled.

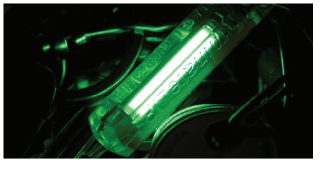
Uses of Ionizing Radiation

The ionization produced by radiation has a wide range of practical applications, from smoke detection and medical imaging to sterilizing medical equipment and generating electricity. The type of radiation used in each application depends on its properties - particularly how it interacts with matter and how deeply it can penetrate different materials.

Safety

Some **smoke detectors** contain a small alpha radiation source that ionizes the air inside the detector. This ionization produces a tiny electric current that is continuously monitored. When smoke enters the detector, the alpha particles are absorbed before they can ionize the air, causing the current to decrease. The detector senses this change and triggers the alarm.

Airport baggage is screened using X rays, allowing potentially dangerous objects to be identified without opening the luggage. Dense materials, such as metals, absorb X rays much more strongly than clothing or plastics, making it easy to distinguish between different objects.



Some specialized **illuminated signs**, used where electricity is unavailable or impractical, contain tritium - a radioactive material that helps produce light continuously for many years without requiring electricity.

Healthcare

Several medical imaging techniques rely on ionizing radiation. Conventional **X ray imaging** and **computed tomography (CT)** both use X rays generated by the imaging system during the procedure. While a conventional X ray produces a single image, a CT scan acquires hundreds of images from different angles, which are then combined to reconstruct a three-dimensional view of the body.

Positron emission tomography (PET) uses a radioactive substance injected into the patient. This technique makes it possible to identify areas of increased metabolic activity, such as many types of tumors.

Magnetic resonance imaging (MRI) uses a phenomenon known as nuclear magnetic resonance to produce detailed images of the inside of the body. Despite the word nuclear in the name, **this procedure does not use ionizing radiation**. Instead, the term refers to the behavior of atomic nuclei placed in a strong magnetic field and exposed to radio waves.

Radiotherapy is used to treat many types of cancer. Depending on the type and location of the tumor, treatment may involve either radioactive sources or particle accelerators. The goal is to damage cancer cells so they can no longer grow and divide.

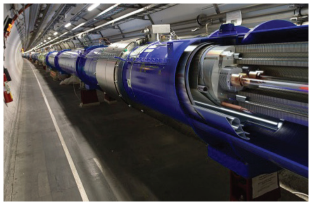
Industry

Nuclear energy is one of the best-known applications of ionizing radiation, but it is far from the only one. Radiation is widely used throughout industry for sterilization, quality control, and non-destructive testing.

Medical equipment, and even certain foods intended for import or export, can be sterilized by exposure to intense sources of ionizing radiation. This destroys microorganisms without heating or damaging the product.

X and gamma rays are also used to inspect metal components and welds, making it possible to detect internal defects without damaging the object being examined. This technique is routinely used to inspect pipelines, storage tanks, bridges, aircraft components, and equipment used in power plants.

Research



Ionizing radiation is an indispensable tool in scientific research. It is used to study the structure of matter, analyze materials, investigate nuclear processes, and monitor the environment. One group of researchers even used cosmic rays to discover an unknown void in the Great Pyramid of Giza.

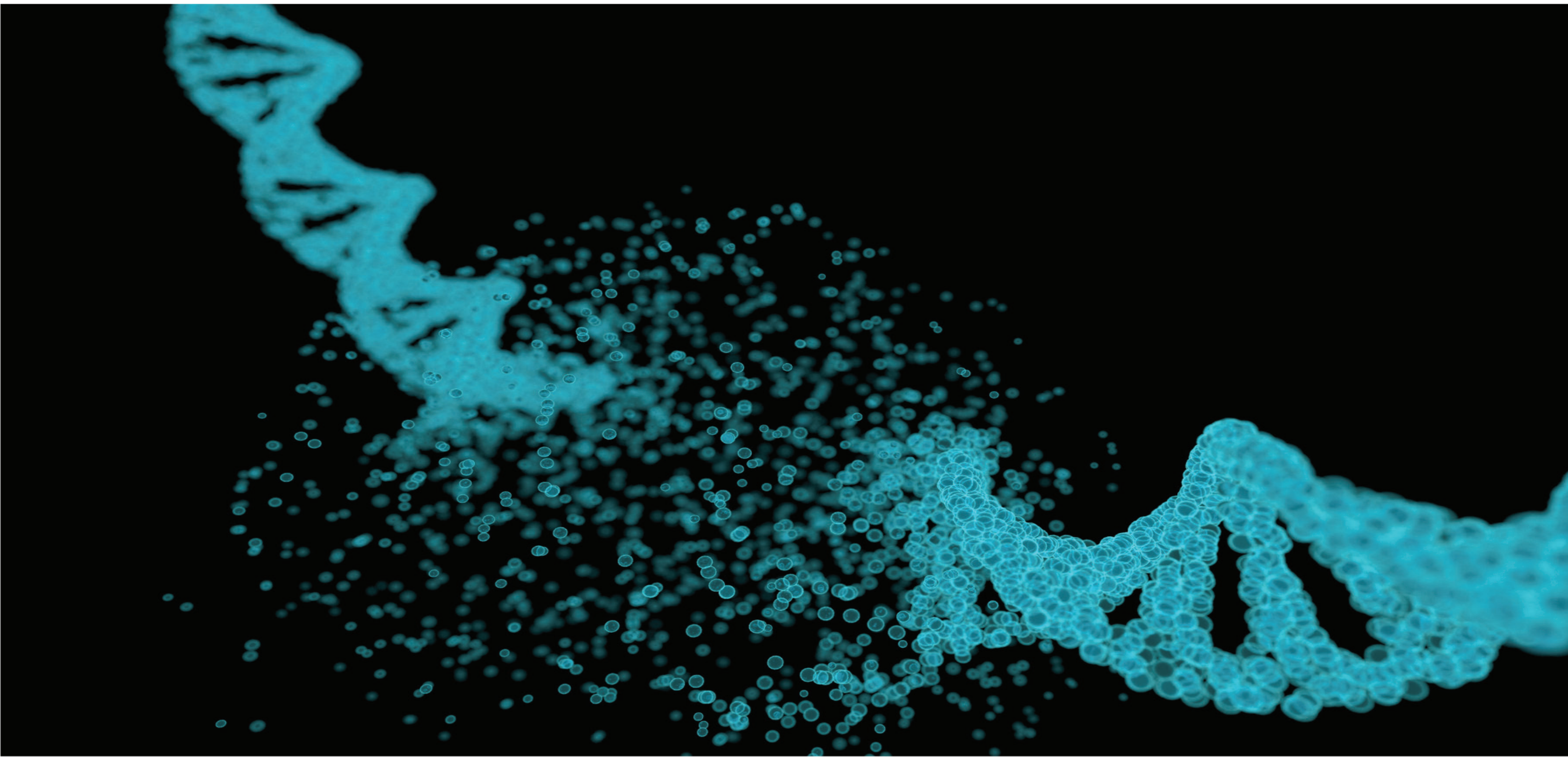
Hazards Associated with Ionizing Radiation

The human body has no natural way of detecting ionizing radiation. Unlike visible light, sound, or heat, ionizing radiation cannot be seen, heard, or felt.

Ionizing radiation transfers energy to matter through ionization. The same process that makes smoke detectors work and allows radiation to be used in cancer treatment can also damage living tissue when the absorbed dose is sufficiently high.

One way in which ionizing radiation produces biological effects is by altering molecules within the body. When ionization removes an electron that is part of a chemical bond, the bond may be broken or its chemical structure altered.

If a DNA molecule is affected, it may suffer damage or even strand breaks. In most cases, however, cells repair this damage successfully using their natural DNA repair mechanisms. In fact, DNA is continuously damaged and repaired as part of normal cellular processes and through exposure to ultraviolet radiation from the Sun. If the damage is too extensive, or if it is repaired incorrectly, it may result in genetic mutations. At very high radiation doses, the amount of damage can exceed the body's ability to repair it.



The effects of exposure to very high radiation doses are well understood and may include burns, damage to internal organs, or acute radiation syndrome. By contrast, the biological effects of low radiation doses are much more difficult to study and remain an active area of research. Several models have been proposed to describe how radiation risk may vary at very low doses. Some assume that any dose carries a certain level of risk, others propose a threshold below which the effects are negligible, while others suggest that very low doses might actually have beneficial effects. These models continue to be investigated. It is important to note that even models assuming some level of risk at any dose do not imply that a small radiation exposure will cause disease. At most, they predict a very small increase in the probability of long-term health effects.

To provide a sense of scale, several examples of effective doses are shown below:

Example	**Effective dose**
Dental X ray	0.005 mSv
Approximately 3-hour flight	0.01 mSv
Computed tomography (CT) scan	5 mSv
Dose at which symptoms of acute radiation syndrome may appear	~400 mSv
Dose at which severe, potentially fatal symptoms may occur	~2000 mSv

These values show that the radiation doses encountered in everyday life, or those used in medical imaging, are far lower than the doses known to cause immediate health effects. Nevertheless, ionizing radiation is always used in a way that keeps exposure as low as reasonably achievable, while still accomplishing the intended purpose.

Detection and Measurements

Ionizing radiation cannot be seen, heard, or felt by the human body. To study it, we rely on instruments capable of detecting and measuring its interactions with matter. At the same time, because radioactive decay is a statistical process, no two measurements are ever exactly the same. Any result must therefore be interpreted with its natural statistical fluctuations in mind.

In this section, you will learn how the RISE detector works, what types of measurements it can perform, and how to interpret the data it produces. These concepts provide the foundation for the laboratory activities presented in the final section of the manual.

Statistical Processes and Background Subtraction

Statistical Processes. Both radioactive decay and the detection of ionizing radiation are statistical processes. This means that the laws of physics cannot predict exactly how many particles will be detected during a given time interval. Instead, they tell us the most probable outcome and how much the result is expected to fluctuate.

Repeating the measurement. If we repeat the same measurement several times under identical conditions, we will obtain slightly different results each time. Even so, the measurements cluster around an average value. As the measurement time increases, the random fluctuations tend to average out, and the result becomes a better estimate of the true value.

Uncertainties. Radiation measurements may be accompanied by uncertainty estimates. Measurement uncertainty provides a mathematically sound way of describing how precise a measurement is and how much confidence we can place in the reported result. These ideas are explored in greater detail in the laboratory activity "The Statistical Nature of Ionizing Radiation".

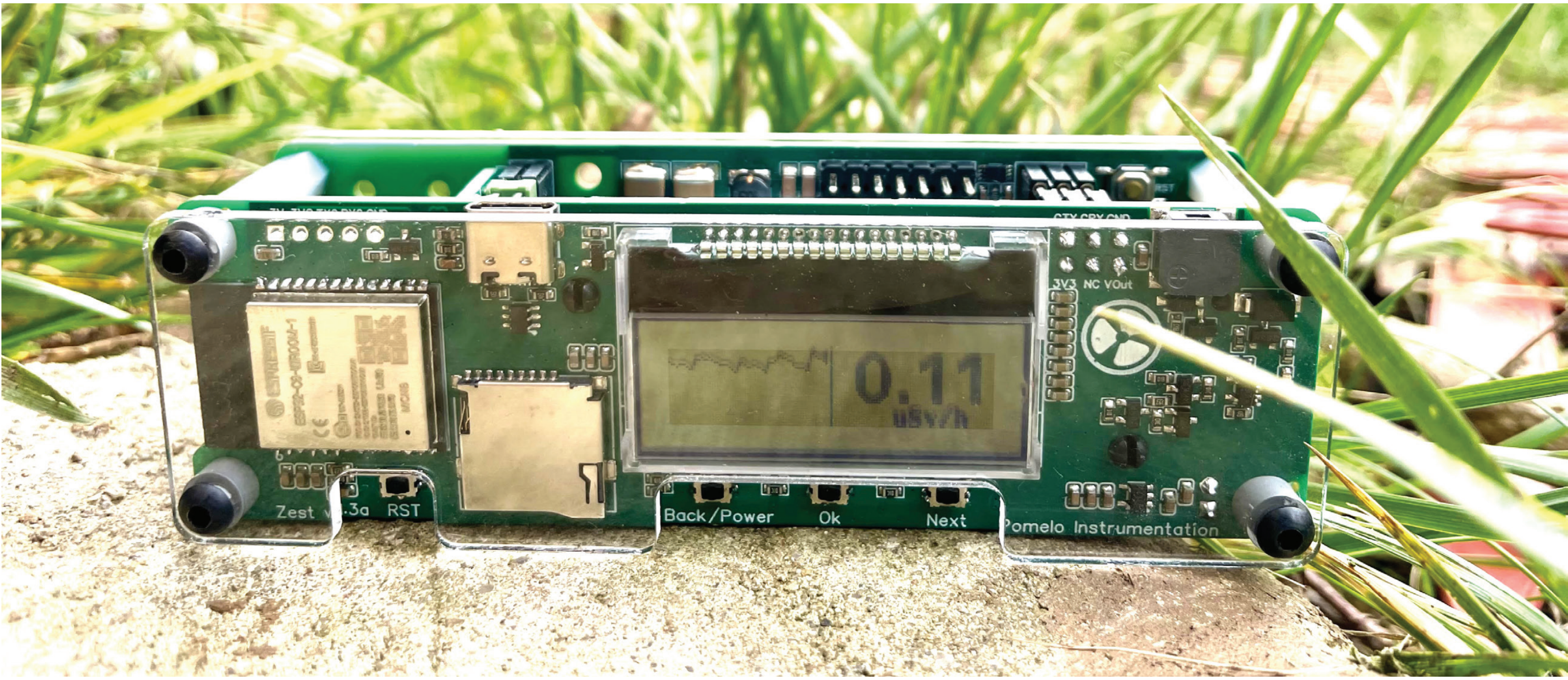
Natural background radiation. All radiation measurements are made in the presence of natural background radiation. This background originates from cosmic radiation, naturally occurring radioactive materials in the soil and in building materials, as well as tiny amounts of naturally occurring radioactive substances within our own bodies.

When measuring the radioactivity of a sample, the detector records both the radiation emitted by the sample and the ever-present background radiation. The measured value is therefore the sum of these two contributions. For this reason, the background radiation is usually measured before the experiment and then subtracted from all subsequent measurements.

For example, suppose the detector measures a natural background of **350 gamma photons per minute**. If placing a sample next to the detector increases the reading to **900 gamma photons per minute**, then the sample itself contributes approximately **550 gamma photons per minute**.

In many situations, this simple subtraction is sufficient. However, if the sample's radioactivity is comparable to the background level, statistical fluctuations may make a weak source impossible to distinguish confidently from the background, or may even produce an apparent difference that is simply due to chance.

In such cases, both the background measurement and the sample measurement must be considered together with their uncertainties. Only then can we judge how convincing the experimental evidence really is. A typical conclusion might therefore be worded as: "**Based on the measurements, there is a 95% probability that this sample is radioactive**."



The RISE Detector

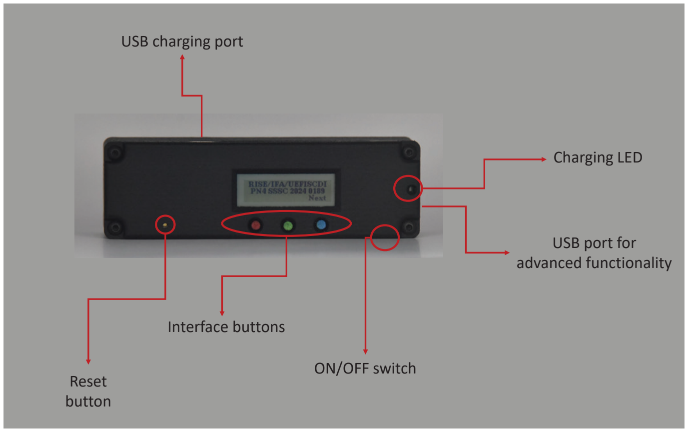
The experiments in this manual use the RISE detector, a gamma-ray detector based on a scintillation crystal and a silicon photomultiplier (SiPM). The detector can measure both the number of gamma rays it detects and the energy of each one.

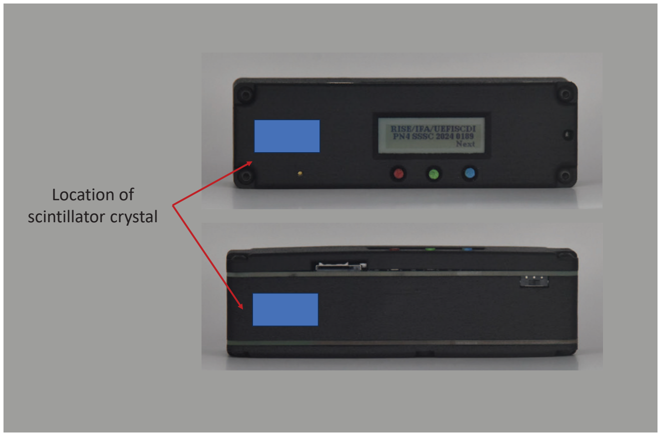
When a gamma ray enters the detector, it interacts with a 2 cm^3 **thallium-doped cesium iodide (CsI(Tl)) scintillation crystal**. This interaction produces a tiny flash of visible light. In general, the amount of light emitted is proportional to the energy deposited by the gamma ray in the crystal.

This flash of light is far too faint to be seen with the naked eye. Instead, it is detected by a **silicon photomultiplier (SiPM)** - an electronic sensor sensitive enough to detect even a single photon of visible light. The SiPM converts the light pulse into an electrical signal.

The detector's electronics measure this electrical signal and determine the energy of the gamma ray that produced it. Since every radioactive isotope emits gamma rays with characteristic energies, measuring these energies allows radioactive materials to be identified.

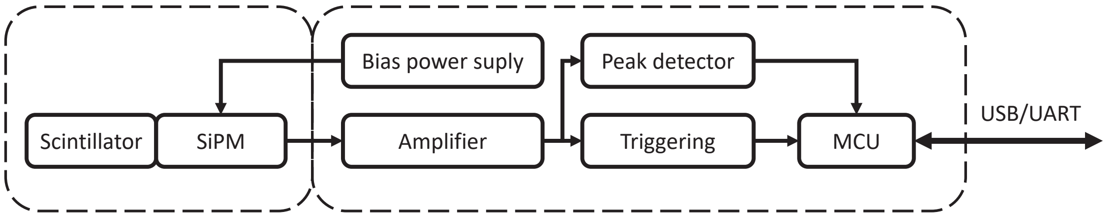
The location of the scintillation crystal inside the detector is shown in the figure below. When measuring weakly radioactive samples, they should be placed as close as possible to this sensitive region to maximize the probability that the emitted gamma rays will reach the detector.





HOW DOES THE DETECTOR WORK?

The block diagram below shows the detector's main functional components and the role of each one.



- **Scintillator** - A CsI(Tl) crystal in which gamma rays produce extremely short flashes of visible light.
- **SiPM (Silicon Photomultiplier)** - Detects the light emitted by the scintillator and converts it into an electrical signal. It is sensitive enough to detect individual photons of light.
- **High-voltage supply** - Provides the approximately 45 V required to operate the silicon photomultiplier.
- **Amplifier** - Increases the amplitude of the electrical signal produced by the SiPM so it can be measured accurately.
- **Peak detector** - The electrical pulses produced by the SiPM last only a few hundred nanoseconds. This circuit temporarily holds the peak value of each pulse, giving the detector electronics enough time to measure it accurately.
- **Trigger** - Detects the arrival of a pulse and signals the microcontroller that a measurement should be performed.
- **MCU (Microcontroller Unit)** - Coordinates the operation of the detector. It measures the electrical pulses, calculates the energy of each detected gamma ray, and transmits the results to a computer or to the detector's user interface.
- **USB / UART** - Communication interfaces used to connect the detector to a computer or other electronic equipment.

Measurement Modes

The RISE detector can display measurement results in several different ways, each designed for a specific purpose. All measurement modes use the same detected gamma rays; the only difference is how the data are processed and presented.

Mode	What does it display?	When is it useful?
CPM	Number of gamma photons detected per minute	Detecting the presence of a radioactive source
μSv/h	Estimated dose rate	Comparing radiation levels in terms of their biological effects
Interval	Number of photons detected during a user-selected time interval	Quantitative experiments with fixed measurement times
Spectrum	Energy distribution of detected gamma photons	Identifying radioisotopes
HTTP	View measurements and control the detector from a computer or smartphone	Detailed spectrum analysis and detector configuration

CPM (Counts Per Minute)

CPM represents the number of gamma photons detected in one minute. The detector updates its measurement every second and reports the corresponding count rate in counts per minute.

Because the number of detected photons relies on a statistical process, the readings in any given time interval fluctuates naturally. To provide a more stable reading, the detector displays a time-averaged value. This reduces the apparent fluctuations but it also causes the displayed reading to respond more slowly to changes in radiation level.

CPM is not a dosimetric quantity because it does not take the energy of the detected photons into account. Instead, it is most useful for detecting radioactive sources and comparing their relative intensities.

μSv/h (Microsieverts per Hour)

The sievert (Sv) is the unit used to quantify the biological effects of ionizing radiation. Since one sievert represents a very large dose, radiation levels are almost always expressed using smaller units such as the millisievert (mSv) or microsievert (μSv).

The value displayed in μSv/h is **an estimate of the dose rate** - the dose that would be received in one hour if the radiation level remained constant..

Interval

In this mode, the detector counts the gamma photons detected during a user-selected measurement interval, ranging from **10 to 600 seconds**.

Use the right button to select the measurement duration, then click the middle button to start the acquisition. During the measurement, the display shows both the remaining time and the number of photons detected so far.

This mode is recommended for experiments that compare measurements collected over the same time interval.

Spectrum

This mode displays the gamma-ray spectrum measured by the detector. Each bar in the histogram represents the number of gamma photons detected within a particular energy range.

Because every radioisotope emits gamma rays with characteristic energies, the spectrum can be used to identify radioactive materials.

The spectrum displayed on the detector itself is intended for quick inspection. For more detailed analysis, it is recommended to connect the detector to a computer or mobile device.

The value shown on the display is the total number of photons accumulated in the spectrum. The **Lin/Log** indicator shows whether the histogram is displayed on a linear or logarithmic scale. Click the middle button to switch between the two display modes, or long press the right button to clear the spectrum and begin a new measurement.

HTTP

This mode allows you to connect to the detector over Wi-Fi and access it from a web browser running on a computer, tablet, or smartphone.

The detector supports two operating modes:

- **Hotspot** – The detector creates its own Wi-Fi network, allowing a computer or mobile device to connect directly
- **Client WiFi** – The detector connects to an existing Wi-Fi network after its network name and password have been configured

Through the web interface, you can:

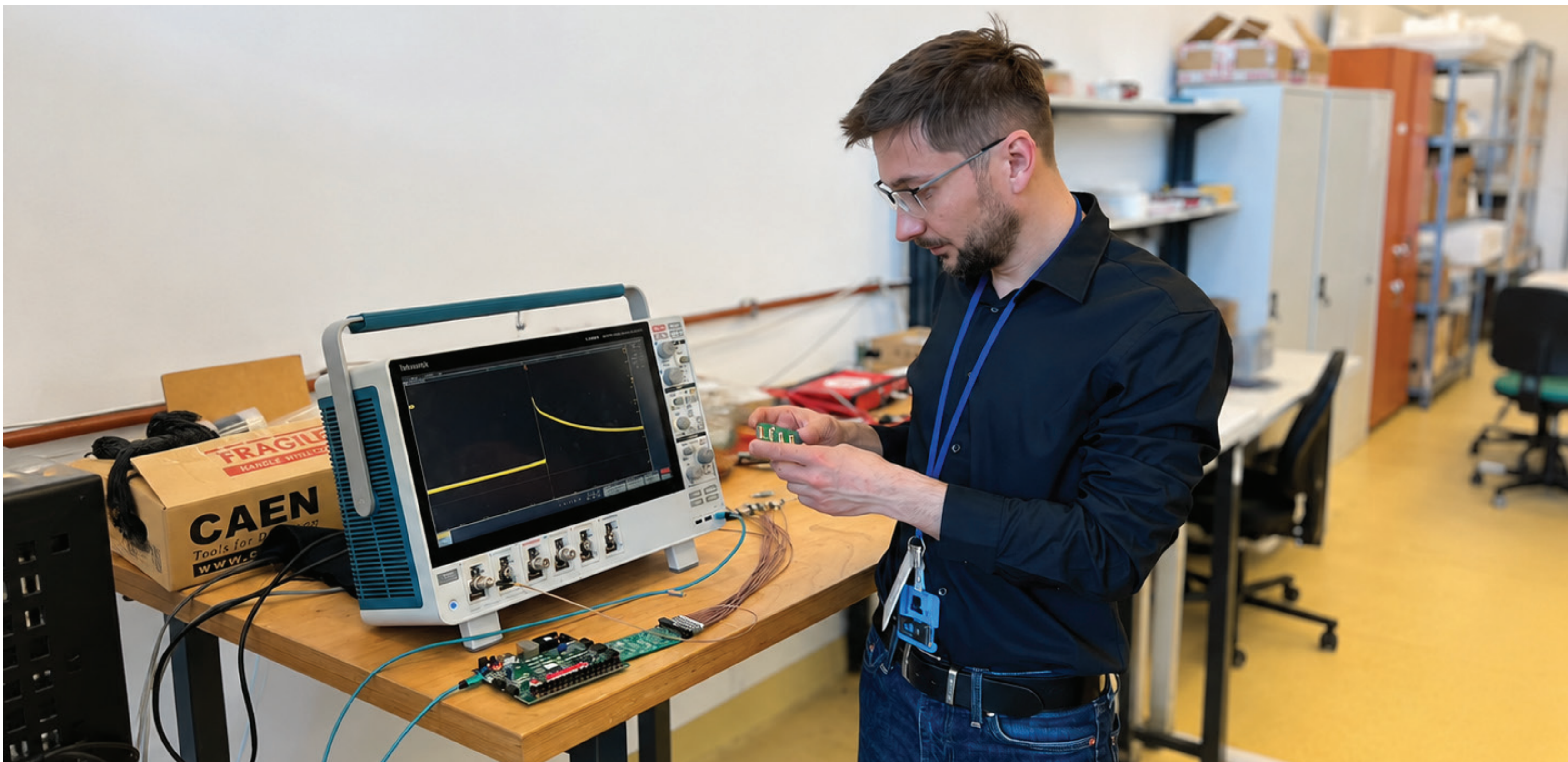
- View the gamma ray spectrum in real time.
- Start and stop data acquisition.
- Download the spectrum in **N42** format, compatible with gamma spectroscopy software such as **InterSpec**.
- Configure the Wi-Fi connection.
- Set the detector's internal clock.

Laboratory Activities

Science advances through observation and experimentation. Now that you have seen what ionizing radiation is and how it can be measured, it's time to put these concepts into practice using the RISE detector.

The laboratory activities in this section are designed to develop both practical measurement skills and a deeper understanding of some of the fundamental concepts of radiation physics, including natural background radiation, shielding, the inverse-square law, and the statistical nature of radioactive decay.

Before starting each experiment, take a moment to predict what you expect to happen. Once you have collected your measurements, compare the results with your prediction and consider what might explain any differences. This process of making predictions, testing them experimentally, and interpreting the results lies at the heart of the scientific method.



Measuring Natural Background Radiation

The goal of this experiment is to measure the natural background radiation at your current location.

Required equipment: RISE detector

Radioactive sources required: None

Description: Our environment contains natural sources of ionizing radiation. These are cosmic radiation, naturally occurring radioactive materials in the soil and in building materials, as well as trace amounts of radioactive substances found within our own bodies. As a result, we are continuously exposed to a low level of radiation known as **natural background radiation**.

The level of background radiation varies from one location to another depending on geographic location, altitude, soil composition, building materials, and other environmental factors. Worldwide, the average annual dose from background radiation is around **2.4 mSv per year** [1].

Measurement:

- Using the RISE detector in the **μSv/h** mode, measure the background radiation dose rate at your current location.
- Calculate the radiation dose that would be accumulated over one year if the measured dose rate remained constant.
- Compare your result with the worldwide average of **2.4 mSv/year.**

Optional: If possible, repeat the measurement at a different location (for example, in a nearby park). Compare the two measurements and note any differences.

Discussion Questions

- Is your measured value higher or lower than the worldwide average? By how much?
- What do you think are the main sources of natural background radiation at the location where you made the measurement?
- Why do you think natural background radiation varies from one region to another?
- Find examples of regions around the world where natural background radiation levels are unusually high. What contributes to these elevated levels?
- If you repeated the measurement a few minutes later, would you expect to obtain exactly the same result? Explain your reasoning.

Radiation Shielding

The goal of this experiment is to observe how the number of detected gamma rays changes when a shielding material is placed between the source and the detector, and how this effect depends on both the shielding material and its thickness.

Required equipment: RISE detector, shielding materials (for example lead, steel, or aluminium).

Radioactive sources required: Preferably a gamma ray lab source (for example ^{241}Am or ^{137}Cs). A naturally radioactive rock may also be used. If neither is available, the experiment can also be carried out using just natural background radiation.

Description: Gamma rays don't always interact with matter, making them much more penetrating than alpha or beta radiation. However, the probability of these interactions increases in dense materials, which is why lead, steel, and concrete are commonly used for gamma ray shielding.

Before taking any measurements: Formulate a hypothesis. How do you think the CPM reading will change if you place a lead plate between the source and the detector? What if you double its thickness? Do you expect all shielding materials to have the same effect?

Option A – Using a Radioactive Source

1. Measure the background count rate (CPM).
2. Keeping the detector in the same position, place the radioactive source a few centimeters away and wait for the reading to stabilize.
3. Determine the contribution of the radioactive source by subtracting the background from the total measurement.
4. Place a shielding material between the source and the detector without changing the position of either.
5. Observe how the CPM reading changes once it has stabilized.
6. Repeat the measurement using different shielding materials or different shielding thicknesses.

Option B – Using Only Natural Background Radiation

1. Measure the background count rate (CPM).
2. Build a shield around the detector's sensitive element, covering as much of its surface as possible with the shielding material.
3. Observe how the CPM reading changes.

Although the effect will be smaller than when using a radioactive source, the underlying phenomenon is the same: shielding materials reduce the probability that gamma rays will reach the detector.

Discussion Questions:

- Which material produced the greatest decrease in the number of detected gamma rays? Why do you think this was the case?
- How did the results change as you increased the thickness of the shielding material?
- Is there any material capable of stopping all gamma rays completely? Explain your answer.
- Why is it important to keep both the source and the detector in the same position throughout the experiment?
- Why is it necessary to subtract the contribution of natural background radiation when using a radioactive source?

The Inverse-Square Law

The goal of this experiment is to verify this law.

Required equipment: RISE detector

Radioactive sources required: Yes

Description: Radioactive materials emit gamma rays equally in all directions. This type of emission is known as **isotropic emission**.

Imagine the radioactive source at the center of a sphere. Every gamma ray emitted by the source passes through the surface of this sphere. If the detector is placed on the sphere's surface, it will intercept only a small fraction of the emitted radiation.

As the distance between the source and the detector increases, the radius of the sphere also increases, and the same amount of radiation is spread over a larger surface area. As a result, an increasingly smaller fraction of the emitted gamma rays reaches the detector.

Mathematically, this relationship is described by **the inverse-square law**: $I \propto \frac{1}{d^2}$ where:

I – radiation intensity;

d – distance between the source and the detector.

Before taking any measurements: Formulate a hypothesis. If you double the distance between the source and the detector, how do you expect the measured intensity to change? Will it be reduced by half, by a factor of four, or by some other amount?

Measurement:

1. Measure the background count rate **(CPM)**.
2. Place the radioactive source at a known distance from the detector and record the CPM reading once it has stabilized.
3. Repeat the measurement for several different source-to-detector distances, changing only the distance between them.
4. For each measurement, calculate the contribution of the radioactive source by subtracting the background count rate.

Complete a table similar to the one below:

d [cm]	$1/d^2$ [cm^{-2}]	Intensity [CPM]	Intensity - background [CPM]
...	...	...	...
...	...	...	...
...	...	...	...

Next, plot **Intensity – Background** as a function of **$1/d^2$.**

If the experimental data points lie approximately on a straight line, the results are consistent with the inverse-square law.

Notice that a graph of intensity as a function of distance is **not** a straight line, whereas a graph of intensity as a function of **$1/d^2$** is. In physics, transformations like this are often used to test whether experimental data follow a particular mathematical relationship.

Discussion Questions:

1. Do your experimental results support the inverse-square law? Which experimental conditions might account for any differences you may observe?
2. Why is it important that the only quantity changed during the experiment is the distance between the source and the detector?
3. Why is it necessary to subtract the contribution of natural background radiation from the measurements?
4. The inverse-square law is at the core of one of the principles of radiation protection: distance. The other two are minimizing exposure time and shielding. Why is increasing the distance such an effective way to reduce radiation exposure?
5. What other physical phenomena follow the inverse-square law?

The Statistical Nature of Ionizing Radiation

The goal of this experiment is to acknowledge these statistical fluctuations and notice how they are influenced by the amount of experimental data collected.

Required equipment: RISE Detector

Radioactive sources required: None

Description: Radioactive decay is a random process: it is impossible to predict exactly when a particular atomic nucleus will decay. However, when we observe a very large number of nuclei, their collective behavior becomes predictable and can be described using statistical methods.

Natural background radiation consists both of the decay of naturally occurring radioisotopes and of other random sources, such as cosmic radiation. As a result, measurements of natural background radiation also exhibit statistical fluctuations.

Before taking any measurements: If the detector remains in the same position and the surrounding conditions do not change, do you expect two consecutive measurements to produce exactly the same result? Explain your reasoning.

Measurement:

1. Using the **Interval** mode, perform **five** measurements of natural background radiation, each lasting **10 seconds**.
2. Write the measured values down.
3. Compare the results. Are all five measurements identical?
4. Calculate the following simple indicator of the variation between the measurements: $\frac{maximum - minimum}{average}$
5. Formulate a hypothesis: if you repeated the experiment using **10-minute** measurements instead of **10-second** measurements, do you think this indicator would be larger, smaller, or approximately the same? If time permits, test your hypothesis experimentally.

Discussion Questions:

1. If two consecutive measurements produce different values, does that mean one of them is incorrect? Explain your answer.
2. Why are the relative statistical fluctuations smaller when the measurement time is longer?

3. The indicator $\frac{maximum - minimum}{average}$ provides a quick way to compare the fluctuations observed in two sets of measurements, but it is not a well-defined quantity. In physics, a mathematically rigorous quantity called the **standard deviation** is used instead:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

where N is the number of measurements, x_i represents the measured values, and μ is their average.

For measurements of ionizing radiation, if only a single measurement containing x detected events is available, the standard deviation can be approximated by

$$\sigma \cong \sqrt{x}$$

This relationship shows that although the absolute fluctuations increase as more events are detected, the relative fluctuations become progressively smaller for longer measurements.

4. Suppose we replace the detector with one that has a much larger scintillation crystal, making it more sensitive and allowing it to detect ten times as many photons during the same measurement interval. If you repeated the 10-second measurements with this detector, how would you expect the observed statistical fluctuations to change?

Optional:

Using the USB interface and the detector's advanced features, you can write a computer program to acquire data from the RISE detector. Automate the data acquisition and record several hundred consecutive measurements of natural background radiation.

Construct a histogram for each of the following measurement intervals:

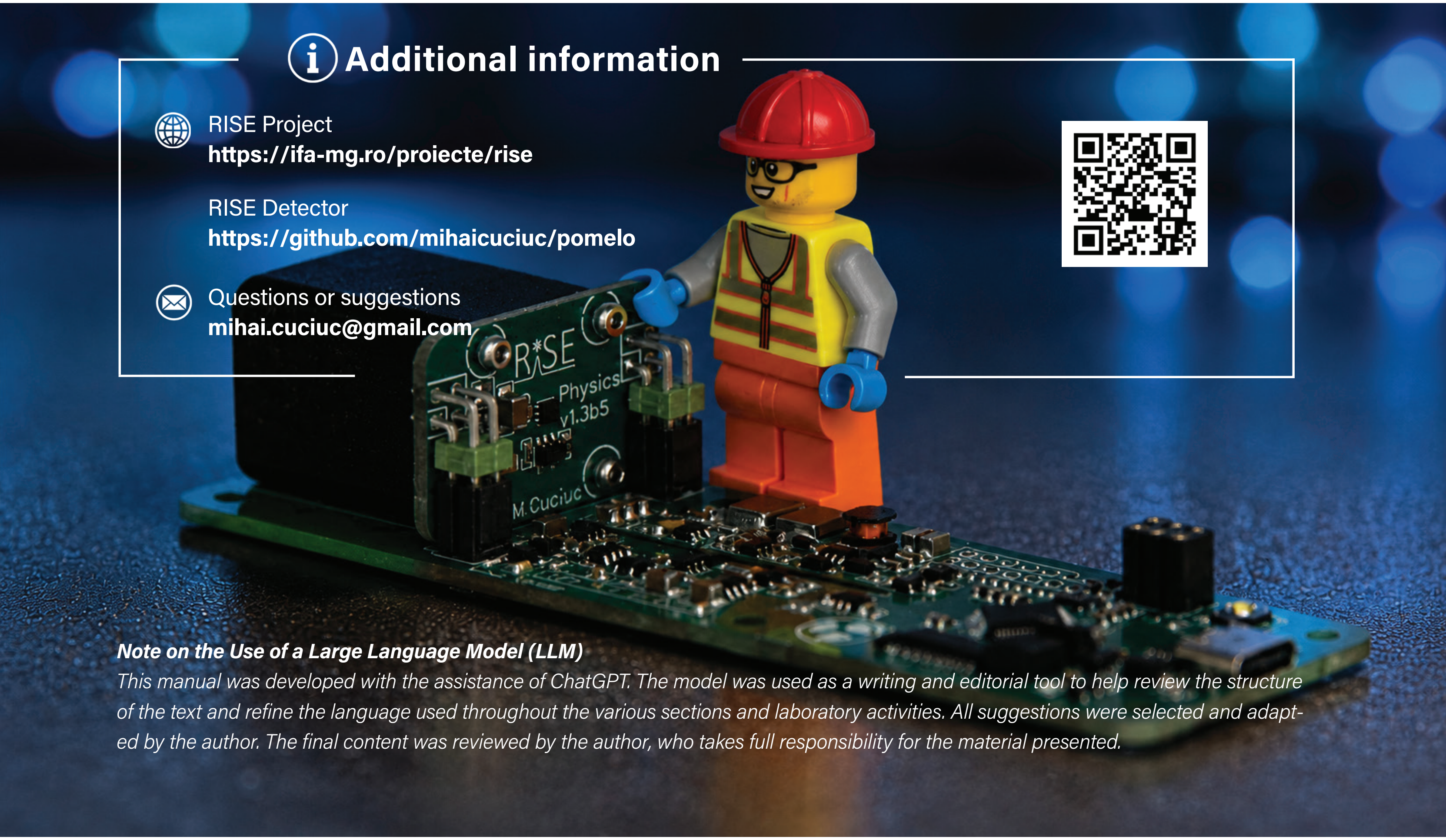
- **1 second**;
- **10 seconds**;
- **100 seconds**.

Compare the resulting histograms. How does the distribution of measured values change as the duration of each measurement increases? What conclusions can you draw about the statistical nature of the radioactive decay process?

References and Contact Information

[1] https://www.unscear.org/unscear/uploads/documents/publications/UNSCEAR_2008_Annex-A-CORR.pdf

[2] https://physics.nist.gov/PhysRefData/XrayMassCoef/tab3.html



Additional information

RISE Project
https://ifa-mg.ro/proiecte/rise

RISE Detector
https://github.com/mihaicuciuc/pomelo

Questions or suggestions
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Note on the Use of a Large Language Model (LLM)

This manual was developed with the assistance of ChatGPT. The model was used as a writing and editorial tool to help review the structure of the text and refine the language used throughout the various sections and laboratory activities. All suggestions were selected and adapted by the author. The final content was reviewed by the author, who takes full responsibility for the material presented.

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